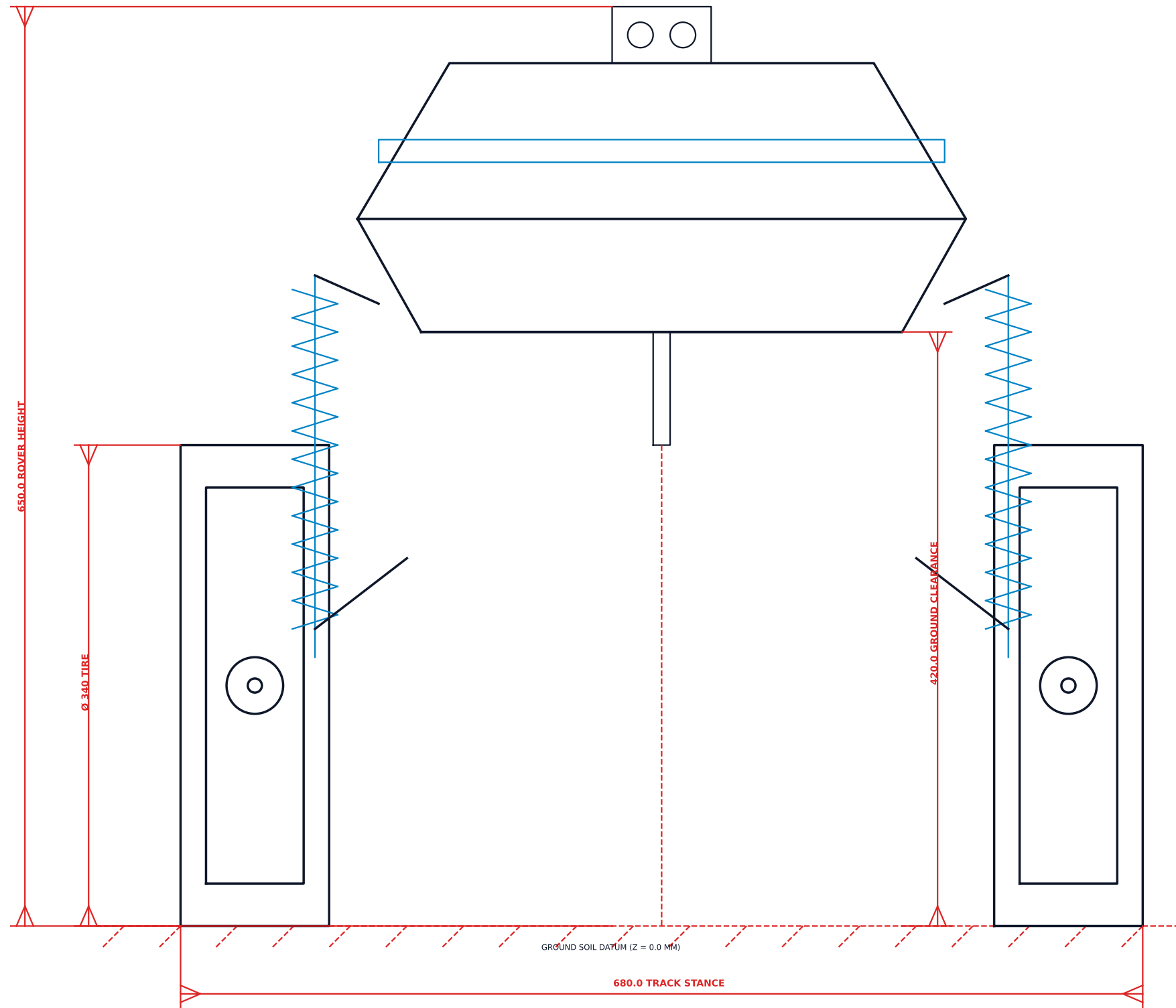


DWG-01

AGRIBOT-X SPECIFICATION - FRONT ELEVATION VIEW



SHEET 1 ENGINEERING NOTES & TOLERANCES (ISO 2768-m):

- Over-canopy cyber strut architecture provides 420.0 mm nominal clearance to straddle crop rows.
- Outer track width (680 mm) designed for standard 700 mm row spacing with zero root compaction.
- Suspension features 4x independent white coilover shock struts with 110 mm active vertical travel.
- Primary Materials: Chassis Al 6061-T6 monocoque, canopy Al 5052-H32, A2-70 metric stainless hardware.

AGRIBOT-X AUTONOMOUS ROVER
TITLE: FRONT ELEVATION VIEW

DESIGN: HIGH-CLEARANCE STRUT
STANDARD: ISO 2768-m TOLERANCES

SCALE: 1:1 TRUE CAD
SHEET: 1 OF 4

DWG NO: AGX-2026-01

REV: C.02 (CLEAN)

DWG-02

AGRIBOT-X SPECIFICATION - TOP PLAN VIEW

750.0 OVERALL LENGTH

380.0 CHASSIS WIDTH

680.0 STANCE WIDTH

520.0 WHEELBASE

SHEET 2 DRIVETRAIN & DOCKING SPECIFICATIONS:

- Drivetrain: 4WD independent BLDC planetary gearmotors (350W each, 1:20 reduction, 19.3 Nm torque).
- Drone Docking Station: Self-centering funnel geometry with spring-loaded gold-plated charging busbars.
- Sensor Turret: Pan/Tilt active tracking head positioned forward of front axle for unobstructed row sweep.
- Beadlock Wheels: CNC machined Al 6061 billet rings with 6x M6 stainless socket fasteners per wheel.

AGRIBOT-X AUTONOMOUS ROVER
TITLE: TOP PLAN VIEW

DESIGN: HIGH-CLEARANCE STRUT
STANDARD: ISO 2768-m TOLERANCES

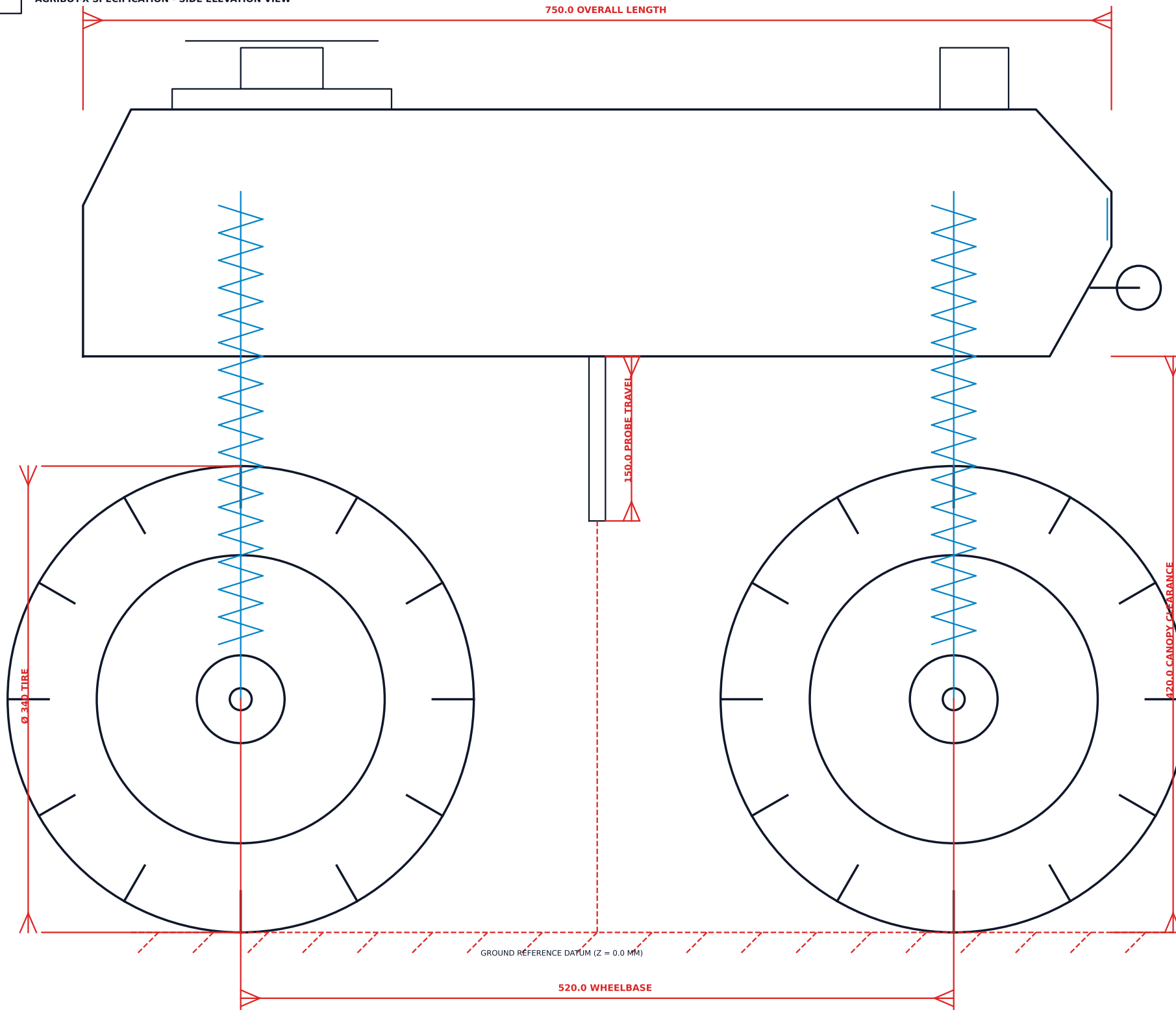
SCALE: 1:1 TRUE CAD
SHEET: 2 OF 4

DWG NO: AGX-2026-02

REV: C.02 (CLEAN)

DWG-03

AGRIBOT-X SPECIFICATION - SIDE ELEVATION VIEW



SHEET 3 KINEMATIC & SOIL PROBE SPECIFICATIONS:

1. Straddle Geometry: 420.0 mm belly clearance guarantees zero canopy friction across arecanut & nursery beds.
2. Soil Penetration: Precision MGN15 linear rail & T8x2 lead screw delivers 150 mm insertion into active root zone.
3. Sensor Cartridge: SS316 multi-parameter electrode measures Moisture, pH, EC, and Temperature via RS485 bus.
4. Center of Gravity: Low-slung battery placement ensures dynamic rollover stability on 25-degree field slopes.

AGRIBOT-X AUTONOMOUS ROVER
TITLE: SIDE ELEVATION VIEW

DESIGN: HIGH-CLEARANCE STRUT
STANDARD: ISO 2768-m TOLERANCES

DWG NO: AGX-2026-03

SCALE: 1:1 TRUE CAD
SHEET: 3 OF 4

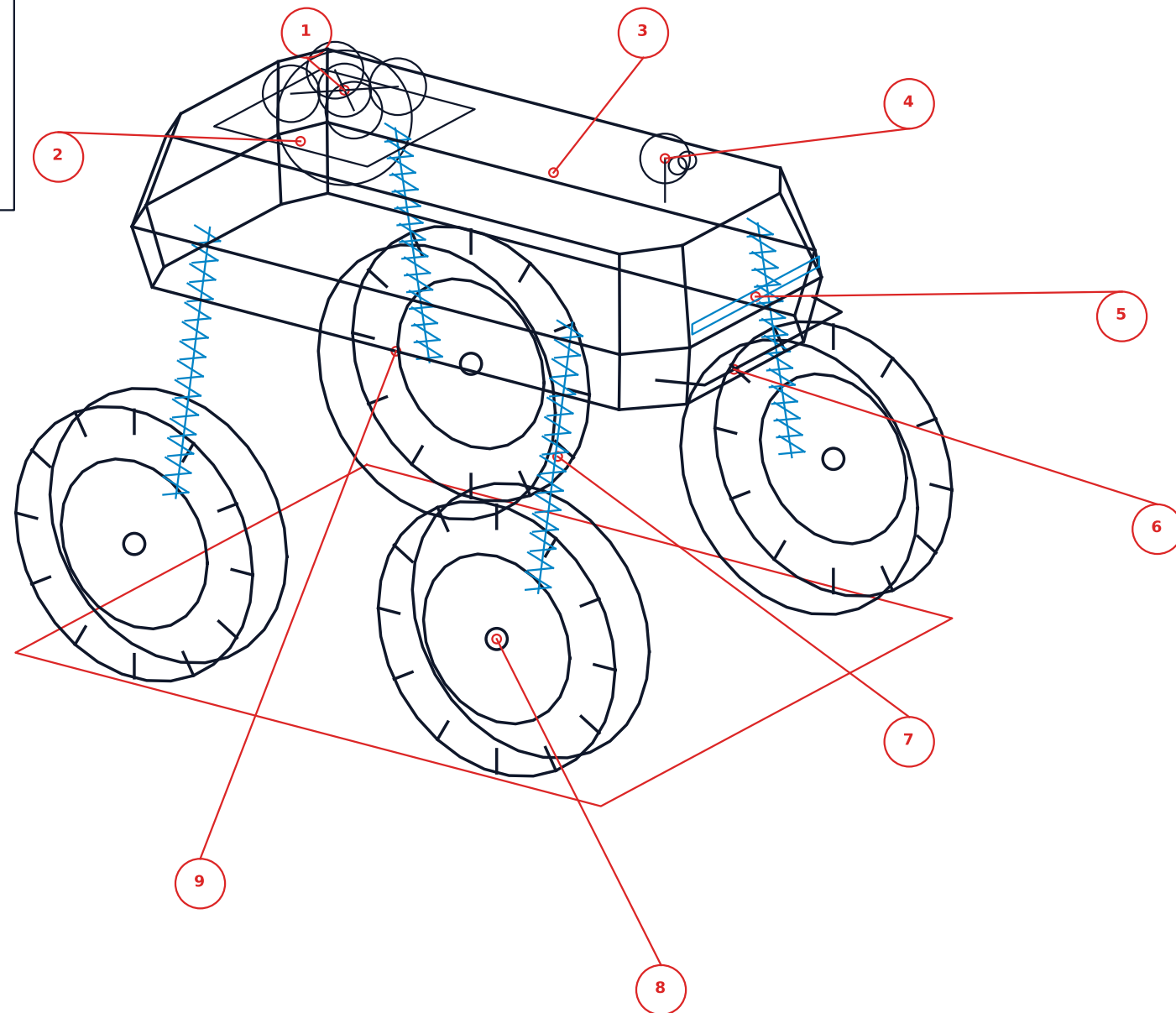
REV: C.02 (CLEAN)

DWG-04

AGRIBOT-X SPECIFICATION - 3D ISOMETRIC ASSEMBLY

ASSEMBLY CALLOUT KEY:

1. Autonomous Surveillance Drone
2. Top Docking & Charging Bay
3. Faceted Titanium Shell Canopy
4. Pan/Tilt Dual-Camera Turret
5. Electric-Blue LED Light Brow
6. Tubular Aluminum Bullbar
7. White Coilover Shock Strut
8. Deep-Lug Tractor Wheel (340mm)
9. Elevated Monocoque Skid Belly



SHEET 4 ASSEMBLY & VERIFICATION GUIDELINES:

1. Torque Specifications: Shock clevis bolts (M8 Grade 8.8) torqued to 22.0 Nm with Loctite 243 blue threadlocker.
2. Spring Preload: Set coilover collar to 15.0 mm initial compression for 32.0 kg nominal vehicle curb weight.
3. Hub Fasteners: 6-lug beadlock pattern torqued in cross-star sequence to 9.5 Nm to ensure airtight tire seal.
4. Enclosure Ingress Protection: Verify continuous silicone O-ring compression before field deployment.

AGRIBOT-X AUTONOMOUS ROVER
TITLE: 3D ISOMETRIC ASSEMBLY

DESIGN: HIGH-CLEARANCE STRUT
STANDARD: ISO 2768-m TOLERANCES

DWG NO: AGX-2026-04

SCALE: 1:1 TRUE CAD
SHEET: 4 OF 4

REV: C.02 (CLEAN)